

Data Scientist/AI Engineer with 6+ years of combined industry and research experience. Specialized in applying deep learning, NLP, Computer Vision, and Generative AI for scalable business impact. Proven ability to deliver insights from machine learning models to marketing stakeholders, applying statistical modeling and data storytelling to real-world use cases. Experienced in customer-centric problem solving, digital (product/predictive) analytics.

EDUCATION

MS (Research)
Electrical & Computer Science Engineering
IIT-Madras, India
2018-2022

B. Tech
Mechanical Engineering
NIT Raipur, India
2013-2017

PUBLICATIONS

High-Speed HDR Video Reconstruction from Hybrid Intensity Frames and Events,
Accepted in the International Conference on Computer Vision and Machine Intelligence(CVMI), 2022 [\[paper\]](#) [\[slides\]](#)

ED.Ai: A Ballistic Simulator for Cardiac and Respiratory Micro-vibration Testing *Accepted in the COMSNETS workshop, 2023 as a Poster* [\[paper\]](#)

Clinical Validation of an Indigenous MicroVibration Vital Parameter Monitor DOZEE VS
Accepted in the NetHealth Workshop, COMSNETS conference, 2023, [\[paper\]](#)

SKILLS

Languages & Frameworks: Python, SQL, PySpark, PyTorch, TensorFlow, Github

AI/ML Expertise: Generative AI, LLM (Transformer Models), Predictive Modelling, Causal Modelling, Deep Learning, Machine Learning, A/B Testing, Time Series Analysis

Cloud, APIs & Developer Tools: Azure Databricks, GCP, BigQuery, FastAPI, Postman

NLP & GenAI: LangChain, Prompt Engineering, Hugging Face, LangGraph, Vectorstore (FAISS, Pinecone)

LANGUAGES

English	9/10
Hindi	9/10

WORK EXPERIENCE

Data Scientist
Tredence Analytics, Bangalore
September, 2023- Present

- Following are the major projects I am involved in:
- Done the migration of 30+ ML pipelines to **Databricks + Spark**, optimizing scalability, cost, & runtime efficiency for a US retail client.
 - Developed and productionized a GenAI-powered CIM summarization solution (RAG + Prompt Chaining + GPT-4o + Pinecone), cutting document generation time by 80%, streamlining team size from 20[→]5, and driving \$4.5M annual savings.
 - Applying machine learning predictive models to patient history data of approximately 2 lakh+ patients for hospital admission/ readmission prediction, ER Risk prediction and hospice condition probability estimation.
 - Given the EMR lab test data, we built RoBERTa based classification model for filtering incorrect lab tests. Then hierarchical clustering algorithm was used to cluster similar lab names that can be integrated with the predictive model for enhanced analysis.
 - Uncertainty Quantification in Neural Networks.
 - Causal Discovery from the structured IID observational data.

Data Scientist
Dozee, Bangalore
June, 2022- August,2023

- The work involves building AI-driven statistical and parametric models to derive meaningful features and information from the complex signal (ballistocardiography and electrocardiography signals). Following were the major projects I was involved in:
- Developed and tested an efficient algorithm to calculate the Respiratory Rate vitals from the Ballistocardiography signals. Testing of data was done on 8000 hrs. of data in uncontrolled settings and 584 hours of data in controlled settings.
 - Analyzing sleep patterns from ballistocardiography(BCG) data for automated sleep staging.
 - Created a pipeline for the analysis of over 14000 Heart Rate (HR) and Respiratory Rate (RR) vital points for getting **US-FDA 510K** clearance.
 - Creating a model for signal quality classification from the ECG data that will be helpful for Arrhythmia
 - Detection pipeline.
 - Apnea/Non-Apnea classification from the BCG signal using the methods of deep learning.

Mentor, Talentsprint & IIT-Madras (Part-Time)
Jul 2023 - Apr 2024

- Advanced Data Science and Machine Learning (ADSML) Program
- Mentored participants in the year-long ADSML program by reviewing programming assignments, resolving doubts, and providing guidance on key machine learning concepts.
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Assisted professionals in capstone projects, offering technical insights and feedback to ensure successful project completion.

Evaluated participant performance, providing constructive feedback to enhance their understanding and practical application of AI/ML techniques.

MAJOR PROJECTS

YouTube Blog Generator with Human Feedback using Agentic AI

Freelance

[\[repository\]](#)

Feb, 2025

- Developed an automated workflow using LangGraph that extracts YouTube transcripts, summarizes video content with an LLM, and generates engaging blog posts.
- Integrated a human-in-the-loop feedback mechanism for iterative content refinement, ensuring high-quality outputs. Designed a Streamlit-based interface to facilitate real-time user review and approval of generated blogs and deployed the model via streamlit.

High-Speed HDR Video Reconstruction using Hybrid Intensity Frames and Events

IIT-Madras, India

[\[slides\]](#)

July, 2021 - April, 2022

- This project aims to prove the efficiency of event-based interpolation as compared to other non-event based techniques for efficient HDR video reconstruction.
- We simulated the event data and used the techniques of deep learning for event-based interpolation and HDR reconstruction.

Curriculum learning in Neural Networks,

IIT Madras, India

[\[report\]](#)

Jan, 2020 - March, 2020

- We demonstrated the concept of curriculum learning on neural networks, where we trained a network initially on the custom-made lines dataset and used their initial layer weights as the initialization for the training on characters dataset.
- We could clearly observe that learning from scratch was much slower as compared to learning from these pre-trained weights.

Ventilator Assignment Problem

Freelance

[\[report\]](#)

December, 2020

- Given patient level data with their characteristics and another data containing time series biometric information of the patients, we aim to reduce patient mortality by identifying high risk patients based on the available data and devoting the limited ventilators to them.
- Firstly, I combined the two data and had done necessary feature engineering, exploratory data analysis and based on feature importance, I had added extra features to give it as an input to various tree-based classifiers. It was found that the gradient boosting technique(Catboost) was found to give better results among all the classifiers.

RECENT HONORS AND AWARDS

- Active contributions to the AI-CoE team at Tredence Analytics resulted in receiving the Founder's Award. [\[Certificate\]](#)
- Received Pat On the Back award from Tredence for contributing in one of the highly influential GenAI project of Document Summarization. [\[Certificate\]](#)